WORKED SOLUTION

STEP I 2001, Question 5

Pure • Differentiating Under the Integral

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q5

The key idea

Both integrals in (i) and (ii) yield to direct integration, the second after the substitution $u = 1 + tx$. The real point is the remark at the end: the integrand of (ii) is the t-derivative of the integrand of (i), and the answers are related in the same way. So differentiating with respect to the parameter t *inside* the integral seems to commute with integrating over x. Doing it once more, and then setting $t = 1$, gives the conjecture.

Solution

Throughout, $t > 0$ and $0 \leq x \leq 1$, so $1 + tx \geq 1 > 0$ and none of the integrands blows up.

Part (i)

Since $\dfrac{\mathrm{d}}{\mathrm{d}x}(1+tx)^{-1} = -t(1+tx)^{-2}$, an antiderivative of $(1+tx)^{-2}$ is $-\dfrac{1}{t(1+tx)}$ (here we use $t \neq 0$). Hence

$$\int_0^1 \frac{1}{(1+tx)^2}\,\mathrm{d}x = \left[-\frac{1}{t(1+tx)}\right]_0^1 = \frac{1}{t}\left(1 - \frac{1}{1+t}\right) = \frac{1}{t}\cdot\frac{t}{1+t} = \frac{1}{1+t},$$

as required.

Part (ii)

The x in the numerator makes the direct approach awkward, so we put $u = 1 + tx$. Then $x = \dfrac{u-1}{t}$, $\mathrm{d}x = \dfrac{\mathrm{d}u}{t}$, and the limits $x = 0, 1$ become $u = 1, 1+t$:

$$\int_0^1 \frac{-2x}{(1+tx)^3}\,\mathrm{d}x = -\frac{2}{t^2}\int_1^{1+t}\frac{u-1}{u^3}\,\mathrm{d}u = -\frac{2}{t^2}\int_1^{1+t}\left(u^{-2}-u^{-3}\right)\mathrm{d}u$$
$$= -\frac{2}{t^2}\left[-\frac{1}{u}+\frac{1}{2u^2}\right]_1^{1+t} = -\frac{2}{t^2}\left(\frac{1}{2}-\frac{1}{1+t}+\frac{1}{2(1+t)^2}\right).$$

Ask yourself How do we tidy the bracket without expanding everything over a common denominator?

Writing $w = \dfrac{1}{1+t}$, the bracket is $\frac{1}{2} - w + \frac{1}{2}w^2 = \frac{1}{2}(1-w)^2$, a perfect square. And $1 - w = \dfrac{t}{1+t}$, so

$$\int_0^1 \frac{-2x}{(1+tx)^3}\,\mathrm{d}x = -\frac{2}{t^2}\cdot\frac{1}{2}\cdot\frac{t^2}{(1+t)^2} = -\frac{1}{(1+t)^2},$$

which is the required result.

The conjecture

Look at the integrands as functions of t with x held fixed:

$$\frac{\partial}{\partial t}(1+tx)^{-2} = -2x(1+tx)^{-3}.$$

So the integrand of (ii) is the t-derivative of the integrand of (i), just as the answer to (ii) is the derivative of the answer to (i). This suggests the rule

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_0^1 f(x,t)\,\mathrm{d}x = \int_0^1 \frac{\partial f}{\partial t}\,\mathrm{d}x.$$

Applying it once more,

$$\frac{\partial}{\partial t}\left(-2x(1+tx)^{-3}\right) = 6x^2(1+tx)^{-4}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\left(-(1+t)^{-2}\right) = 2(1+t)^{-3},$$

so we conjecture that, for $t > 0$,

$$\int_0^1 \frac{6x^2}{(1+tx)^4}\,\mathrm{d}x = \frac{2}{(1+t)^3}.$$

Putting $t = 1$, the conjectured value is

$$\int_0^1 \frac{6x^2}{(1+x)^4}\,\mathrm{d}x = \frac{2}{8} = \frac{1}{4}.$$

Checks

- At $t \to 0$ the integrand in (i) tends to 1 and the answer $\frac{1}{1+t}$ tends to 1; in (ii) the integrand tends to $-2x$, whose integral is -1, matching $-\frac{1}{(1+t)^2} \to -1$.
- We weren't asked to verify the conjecture, but it is quick to do: with $u = 1+x$ the integral becomes $6\int_1^2 (u^{-2} - 2u^{-3} + u^{-4})\,\mathrm{d}u = 6\left[-\frac{1}{u} + \frac{1}{u^2} - \frac{1}{3u^3}\right]_1^2 = 6\left(-\frac{7}{24} + \frac{1}{3}\right) = \frac{1}{4}$.
- Simpson's rule gives 0.250000 for the final integral, and agrees with $\frac{2}{(1+t)^3}$ to nine decimal places for 25 different values of t.

Common slips

- Forgetting the factor $\frac{1}{t}$ from $\mathrm{d}x = \mathrm{d}u/t$ in (ii): there are *two* factors of $\frac{1}{t}$, one from x and one from $\mathrm{d}x$.
- Not changing the limits to $u = 1$ and $u = 1 + t$ after substituting.
- Differentiating $-(1+t)^{-2}$ and getting $-2(1+t)^{-3}$: the two minus signs cancel, so the conjectured value is positive (as it must be, since the integrand is positive).

Beyond STEP

The rule we conjectured is *differentiation under the integral sign* (Leibniz's rule). It is a theorem of real analysis that it holds when f and $\partial f/\partial t$ are continuous on a closed rectangle, which is true here for $0 \le x \le 1$ and t in any interval $[a, b]$ with $a > 0$. Repeating the argument gives $\int_0^1 \frac{x^n}{(1+tx)^{n+2}}\,\mathrm{d}x = \frac{1}{(n+1)(1+t)^{n+1}}$ for every n. Richard Feynman famously used this trick to evaluate integrals that resisted other methods. For the theory behind swapping limits, derivatives and integrals, see Bartle and Sherbert, *Introduction to Real Analysis*.

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